Lecture 6

The Euclidean path integral gives a direct test of whether pure three-dimensional gravity has the spectrum of a two-dimensional conformal field theory. Lecture 6 constructs the sum over smooth solid tori, derives its one-loop factor both from asymptotic symmetry and from heat kernels, and extracts the spectrum of the regularized sum. Negative spectral weights and a continuous energy distribution identify the limitations of this candidate definition of quantum gravity.

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Conventions

We retain $\Lambda = -1/\ell^2$, $[G] = [\ell]$, and the spatial boundary circle of circumference $2\pi\ell$. Except in the explicitly supersymmetric examples, the theory is bosonic Einstein gravity with equal central charges

$$c_L = c_R = c = \frac{3\ell}{2G}, \qquad k_{\mathrm{CS}} = \frac{\ell}{4G} = \frac{c}{6}, \qquad k \equiv \frac{c}{24} = \frac{k_{\mathrm{CS}}}{4}. \tag{1}$$

Here k_{CS} is the fundamental-trace level used in lecture 2; k uses the vector-trace normalization of [1]. Their factor of four is a trace convention, not a quantum correction. Until §1.1 (lecture 5), k is just a real parameter. The relation to ℓ/G is the classical Brown–Henneaux normalization. In the quantum characters below, c denotes the physical central charge; any local renormalization of the coupling and vacuum energy is absorbed into that parameter. This convention does not specify a universal shift between a bare Chern–Simons level and c. In lecture 6 we impose no integrality assumption on c. [9][8, §1, footnote 1]

Write $h, \bar{h}$ for the plane eigenvalues of $L_0, \bar{L}_0$, so the vacuum has $(h, \bar{h}) = (0, 0)$. The cylinder energies and angular momentum are

$$E_L = h - \frac{c}{24} = \frac{\ell M + J}{2}, \qquad E_R = \bar{h} - \frac{c}{24} = \frac{\ell M - J}{2}. \tag{2}$$

Thus $\ell M = h + \bar{h} - c/12$ and $J = h - \bar{h}$; this is the orientation convention of lecture 3. Some references reverse the sign of J, and Witten often denotes the shifted cylinder generator itself by L_0. We translate both conventions into (2). [1, §3.1][7, §1]

For a real Euclidean boundary torus,

$$w \sim w + 2\pi \sim w + 2\pi\tau, \quad \tau = x + iy, \quad y > 0, \quad q = e^{2\pi i\tau}, \quad \bar{q} = e^{-2\pi i\bar{\tau}}, \quad \bar{\tau} = \tau^*. \tag{3}$$

The trace is

$$Z(\tau, \bar{\tau}) = \mathrm{Tr}_{\mathcal{H}} q^{E_L} \bar{q}^{E_R} = \mathrm{Tr}_{\mathcal{H}} e^{-2\pi y \ell M + 2\pi i x J}. \tag{4}$$

Here $\widehat{\beta} = \beta_{\mathrm{phys}}/\ell = 2\pi y$ and $2\pi x$ is a real angular twist. A real Lorentzian angular potential is obtained by analytic continuation: $\beta_L = \widehat{\beta}(1 - \widehat{\Omega})$, $\beta_R = \widehat{\beta}(1 + \widehat{\Omega})$, with $\widehat{\Omega} = \ell\Omega$, $\tau = i\beta_L/(2\pi)$ and $\bar{\tau} = -i\beta_R/(2\pi)$. Then τ and $\bar{\tau}$ need not be complex conjugates. This continuation is already needed for ordinary rotating thermodynamics; it should be distinguished from enlarging the set of bulk configurations. [6, §2] [checked]

1 The Euclidean Einstein path integral with torus boundary

1.1 The action, the observable, and the choice of configurations

We now use the bulk action to construct a candidate partition function, without assuming a microscopic CFT answer. In lecture 4 we established the classical thermal weights of AdS and BTZ. The additional task is to specify which Euclidean fillings contribute, include their quantum fluctuations, and test whether their sum can count states. Unlike the protected indices of lecture 5, this is an ordinary thermal observable, so a CFT interpretation requires a positive spectral measure. [6, §§1–2][7, §1]

With the outward-pointing normal and the conventions of lecture 4, the regulated Euclidean Einstein action is

$$I_E = -\frac{1}{16\pi G}\int_M \sqrt{g}\left(R + \frac{2}{\ell^2}\right)\mathrm{d}^3x - \frac{1}{8\pi G}\int_{\partial M}\sqrt{\gamma}K\,\mathrm{d}^2x + \frac{1}{8\pi G\ell}\int_{\partial M}\sqrt{\gamma}\,\mathrm{d}^2x. \tag{5}$$

Here γ is the induced metric at a radial cutoff and K is the trace of its extrinsic curvature. The second term makes the variational problem compatible with fixing the boundary metric; the counterterm in the third removes the leading volume divergence. The cutoff is removed after combining these terms. For the flat boundary torus there is no integrated curvature anomaly term. [13][6, §2.2]

Fix the conformal boundary to be the torus with modulus τ. The path integral should include all allowed interiors compatible with that same boundary data, weighted by this action:

$$Z_{\text{grav}}(\tau,\bar{\tau}) = \sum_{M:\,\partial_\infty M = T^2} \int_{\mathcal{C}_M} \frac{\mathcal{D}g}{\text{Diff}_0}\, e^{-I_E[g]}. \tag{6}$$

The sum labels bulk manifolds M, $\partial_\infty M$ denotes their conformal infinity, and $\mathcal{C}_M$ is the integration contour for metrics on each manifold. The quotient removes redundant diffeomorphisms acting trivially on the fixed boundary data. In particular it does not remove the asymptotic symmetry excitations that carry the boundary charges. Equation (6) is a formal definition until its saddle set, contour, and regularization are supplied. [6, §§1–2]

Even this formal expression contains a choice not fixed by writing the Einstein action: the Euclidean conformal mode has a wrong-sign kinetic term, so integration over all real Euclidean metrics does not converge. Perturbation theory uses a contour rotation for that mode. The nonperturbative contour, and which additional topologies or complex saddles belong to it, are not determined by the Gaussian computation. [8, §2.3][6, §§1,4]

The candidate we will evaluate sums the perturbative contributions about smooth, real, orientable, complete hyperbolic fillings with the prescribed single torus conformal boundary and no additional cusp ends. These restrictions give a concrete saddle set, but do not establish that every contribution to the formal expression (6) has been included. The distinction matters when the candidate fails the Hilbert-space test. [6, §§1–2]

For an ordinary compact unitary CFT with a unique vacuum, the answer must admit

$$Z(\tau,\bar{\tau}) = \sum_\alpha d_\alpha q^{h_\alpha - c/24}\bar{q}^{\bar{h}_\alpha - c/24}, \qquad d_\alpha \in \mathbb{Z}_{\geq 0}, \quad h_\alpha, \bar{h}_\alpha \geq 0, \quad h_\alpha - \bar{h}_\alpha \in \mathbb{Z}. \tag{7}$$

There is a unique identity state, and a discrete spectrum with finite multiplicities below any fixed energy. The dimensions need not be integers: discreteness of the energies is different from integrality of their multiplicities. Modular invariance alone imposes neither positivity nor discreteness. [6, §2.3][7, §1]

1.2 Classifying and distinguishing the solid tori

To turn the path integral into a saddle sum, we must distinguish two questions: what topology can a smooth saddle have, and how is its boundary attached to the specified torus? For the saddle set above, the topology is always a solid torus, $D^2 \times S^1$. Nevertheless, there are infinitely many fillings of a *marked* boundary torus, one with specified spatial and thermal cycles A, B. Their difference is which boundary cycle becomes contractible in the interior. We will first see this in the thermal metric, then derive the integer labels that count the possibilities. [6, §2.1]

The thermal-AdS metric is

$$\mathrm{d}s^2 = \ell^2\left[\mathrm{d}\rho^2 + \cosh^2\rho\,\mathrm{d}t_E^2 + \sinh^2\rho\,\mathrm{d}\varphi^2\right], \qquad (t_E,\varphi) \sim (t_E + 2\pi y, \varphi + 2\pi x), \quad \varphi \sim \varphi + 2\pi. \tag{8}$$

Near $\rho = 0$, the (ρ,φ) directions form a smooth disk: $\mathrm{d}\rho^2 + \rho^2\mathrm{d}\varphi^2$. Its boundary is the spatial cycle A, which shrinks to a point at the center. The Euclidean-time identification leaves a noncontractible circle there. In the nonrotating BTZ filling the roles are reversed: Euclidean time is the

polar angle of a disk that closes at the horizon, while the spatial circle remains noncontractible. Both interiors are solid tori, but they fill different cycles of the same boundary. Figure 1 shows this distinction without using a coordinate-dependent embedding. [6, §2.1] [checked]

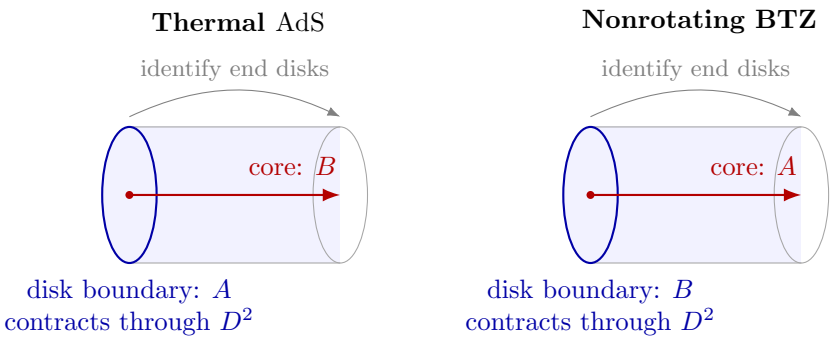


Figure 1: A solid torus cut open along a meridional disk, drawn as $D^2 \times [0,1]$. Gluing the end disks closes the red interval into the noncontractible core circle. The blue boundary of either disk is the contractible meridian. The topology is the same in both panels; the boundary labels distinguish the fillings. The figure takes $x = 0$ so the gluing has no angular twist.

A general filling makes $C = dA + rB$ contractible. The integers must be coprime: a nonprimitive cycle winds several times around a simpler closed curve and cannot be the simple meridian of a smooth solid torus. A second cycle $L = bA + aB$ must intersect C once, giving $ad - br = 1$. Its period relative to the meridian is consequently

$$\tau_\gamma = \frac{a\tau + b}{r\tau + d}, \qquad \gamma = \begin{pmatrix} a & b \\ r & d \end{pmatrix} \in \Gamma_\infty \backslash \mathrm{PSL}(2,\mathbb{Z}). \tag{9}$$

This is a change to the basis in which the selected meridian plays the role of the spatial cycle of thermal AdS. Evaluating its thermal answer at τ_γ therefore gives the answer for this filling in the original boundary coordinates. [6, §2.1][3, §1] [checked]

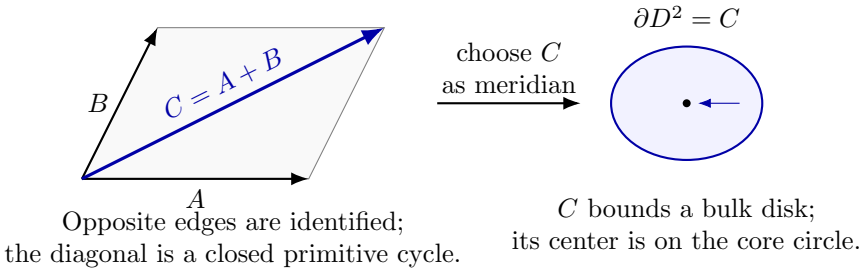


Figure 2: A different meridian on the marked boundary torus. The example $(r, d) = (1, 1)$ makes $A + B$ contractible. Its complementary cycle is defined only up to addition of this meridian.

There are two redundancies in this labeling. Reversing both r, d reverses the orientation of the meridian and gives the same filling; this accounts for using PSL rather than SL. Also, replacing L by $L + nC$ does not change which cycle bounds the disk. It sends γ to $T^n\gamma$ and is removed by the left quotient by $\Gamma_\infty = \langle T \rangle$. Representatives can thus be chosen as $(r, d) = (0, 1)$ and all $r > 0$, $d \in \mathbb{Z}$ with $(r, d) = 1$. In particular, $(0, 1)$ is thermal AdS and $(1, 0)$ is its S image, BTZ. We reserve c for the central charge and use r for the lower-left matrix entry. [6, §2.1] [checked]

To justify that this construction exhausts the stipulated smooth saddles, use the Einstein equations. In three dimensions $R_{\mu\nu} = -2g_{\mu\nu}/\ell^2$ fixes the full Riemann tensor, so the metric has constant negative curvature. A complete saddle is a quotient of hyperbolic space, whose upper-half-space metric is

$$\mathrm{d}s^2 = \ell^2 \frac{\mathrm{d}u^2 + \mathrm{d}z\mathrm{d}\bar{z}}{u^2}, \qquad u > 0. \tag{10}$$

Its orientation-preserving isometries form $\mathrm{PSL}(2,\mathbb{C})$. With one torus conformal boundary and no extra cusp ends, the smooth quotients are cyclic loxodromic quotients. A loxodromic isometry translates along a geodesic and may rotate around it. For thermal AdS a generator is

$$(u, z) \longmapsto (e^{2\pi y} u, e^{2\pi y - 2\pi i x} z). \tag{11}$$

Writing $z = e^{-iw}$ gives the boundary periods 2π and $2\pi\tau$. The quotient closes the translation axis into the core circle of the solid torus. This realizes geometrically the topology and the gluing drawn above. [6, §2.1] [checked]

The assumptions exclude other quotients for identifiable reasons. Parabolic identifications $z \mapsto z + m + n\tau$ also give a torus at conformal infinity, but leave a cusp end as $u \to \infty$, where its cycles collapse. Elliptic identifications have fixed points and would introduce orbifold singularities. Neither belongs to the smooth, no-cusp saddle set chosen in §3.1. [6, §2.1]

Finally, a boundary modular transformation permutes the meridians in this list. It need not extend to a diffeomorphism of a *fixed* filling: a map that exchanges its contractible and noncontractible cycles cannot extend smoothly over its disk. Thus the boundary symmetry permutes distinct bulk contributions, rather than allowing us to discard them as repeated descriptions of a single marked filling. Summing a set closed under this permutation is what will make the boundary observable modular invariant. [6, §§2.1,3]

1.3 Classical weights and the Hawking–Page transition

Global AdS_3 has $M = -1/(8G)$ and $J = 0$, so its thermal action is $\beta_{\mathrm{phys}} M = -\widehat{\beta} c/12$. Equivalently the regulated bulk action (5) gives

$$I_{\mathrm{cl}}(\tau) = 2\pi i k(\tau - \bar{\tau}) = -4\pi k y = -\frac{\pi c}{6} y, \qquad Z_{\mathrm{cl}} = |q|^{-2k}. \tag{12}$$

The action is negative; its Boltzmann factor grows at low temperature because the vacuum cylinder energy is negative. For a general filling,

$$I_\gamma = -4\pi k \, \mathrm{Im}(\gamma\tau) = -\frac{\pi c}{6} \frac{y}{|r\tau + d|^2}. \tag{13}$$

The equality uses $\mathrm{Im}(\gamma\tau) = y/|r\tau + d|^2$, obtained directly from $ad - br = 1$. [6, §2.2] [checked]

For the S image the answer is $I_S = -4\pi k y/|\tau|^2$. To keep rotation before specializing to $J = 0$, analytically continue to the independent β_L, β_R defined above. Then

$$\log Z_{\mathrm{cl}}^{\mathrm{BTZ}} = 4\pi^2 k \left(\frac{1}{\beta_L} + \frac{1}{\beta_R} \right), \qquad E_L = -\partial_{\beta_L} \log Z = \frac{4\pi^2 k}{\beta_L^2}, \quad E_R = \frac{4\pi^2 k}{\beta_R^2}. \tag{14}$$

Using $\beta_L = 2\pi\ell/(r_+ + r_-)$ and $\beta_R = 2\pi\ell/(r_+ - r_-)$ reproduces $E_{L,R} = (r_+ \pm r_-)^2/(16G\ell)$. The thermodynamic entropy is

$$S = \log Z + \beta_L E_L + \beta_R E_R = 8\pi^2 k \left(\frac{1}{\beta_L} + \frac{1}{\beta_R} \right) = \frac{2\pi r_+}{4G}. \tag{15}$$

This is a direct check of the normalization of the general modular weight, not an independent assumption about the microscopic spectrum. [6, §2.2] [checked]

At $x = 0$, $y = \widehat{\beta}/(2\pi)$, the two actions are

$$I_{\rm AdS} = -\frac{c\widehat{\beta}}{12}, \qquad I_{\rm BTZ} = -\frac{\pi^2 c}{3\widehat{\beta}}. \tag{16}$$

They cross at $\widehat{\beta} = 2\pi$, the result of lecture 4. Thermal AdS dominates for $y > 1$ and BTZ for $y < 1$. More generally, the classically preferred filling minimizes $|r\tau + d|$. In the standard fundamental domain $|x| \le 1/2$, $|\tau| \ge 1$, this is thermal AdS; its modular images give the other dominance regions. Sharp phase boundaries describe the large-c saddle approximation, not a generic finite-c nonanalyticity of a compact CFT trace. [6, §6][3, §5.3]

Exercise 1. Show that two fillings (r, d) and (r', d') have equal classical action on $|r\tau + d| = |r'\tau + d'|$. Verify the thermal-AdS/BTZ boundary $|\tau| = 1$ and show directly that all other primitive pairs have $|r\tau + d| \ge 1$ in the standard fundamental domain. [checked]

2 The contribution of one filling

2.1 Boundary gravitons from the asymptotic symmetry algebra

The classical action determines the leading exponential of a saddle, but an exact state-counting test is sensitive to its order-one prefactor. In three dimensions that prefactor cannot be obtained by counting local graviton polarizations, since there are none. The states instead come from asymptotic transformations that carry boundary charges. As in lecture 5, their representation under the Virasoro algebra determines the perturbative spectrum. [6, §2.2]

Brown–Henneaux transformations with nonzero boundary charges are not redundant gauge transformations. Acting on the AdS vacuum, their Virasoro generators create physical boundary gravitons. In the perturbative vacuum sector, the states are

$$\prod_{n=2}^{\infty} L_{-n}^{N_n} \prod_{m=2}^{\infty} \bar{L}_{-m}^{\bar{N}_m} |0\rangle, \qquad N_n, \bar{N}_m \in \mathbb{Z}_{\ge 0}, \tag{17}$$

with finite occupation at any finite energy. Equations (10)–(11) (lecture 5) then give

$$Z_{\rm vac}(\tau, \bar{\tau}) = |\chi_{\rm vac}(\tau)|^2 = |q|^{-c/12} \prod_{n=2}^{\infty} \frac{1}{|1 - q^n|^2}. \tag{18}$$

The factor $|q|^{-c/12}$ is the classical action; the product is the one-loop partition function. Omitting the $n = 1$ mode expresses the vacuum's invariance under the global conformal subgroup. Including it would count a null vector as a physical state. [9][6, §2.2]

A useful check comes from the norm of one excitation. Using $L_n^\dagger = L_{-n}$ and (10) (lecture 5),

$$\langle 0|L_n L_{-n}|0\rangle = \frac{c}{12} n(n^2 - 1), \qquad n \ge 2. \tag{19}$$

It is positive for $c > 0$, while the $n = 1$ expression vanishes. At the $c > 1$ values under consideration the vacuum module has its usual unitary Virasoro structure, consistent with interpreting these descendants as physical excitations of the vacuum. [6, §2.2] [checked]

The symmetry argument gives more than oscillator counting at large c. The commutator $[L_0, L_{-n}] = nL_{-n}$ fixes the exact spacing of levels, while the vacuum representation fixes their degeneracies. Continuous perturbative corrections cannot shift those levels without changing the representation. With c understood as the physical central charge, this fixes the contribution of the perturbative vacuum sector to (18); its logarithm has a classical term and a one-loop term, with no further $1/c$ series. This one-loop exactness assumes that perturbative states remain precisely the vacuum module, and it leaves room for new nonperturbative sectors. A one-loop determinant calculation alone does not prove that no such sectors exist. [6, §2.2]

2.2 Gauge fixing and the heat-kernel method

The symmetry argument predicts the entire vacuum contribution. A direct Gaussian path integral on $\mathbb{H}^3/\mathbb{Z}$ tests how that answer emerges from bulk fluctuations and gauge fixing. Although there is no propagating graviton, the cancellation of local gauge modes leaves a nontrivial contribution from the quotient and boundary conditions. [8, §§2–4]

Set $\ell = 1$ temporarily, so the proper-time variable below is measured in units of ℓ^2. Let $g_{\mu\nu} = g^{(0)}_{\mu\nu} + h_{\mu\nu}$. The de Donder gauge is

$$\nabla^\mu \left(h_{\mu\nu} - \tfrac{1}{2} g^{(0)}_{\mu\nu} h \right) = 0, \qquad h = g^{(0)\mu\nu} h_{\mu\nu}. \tag{20}$$

Separate $h_{\mu\nu}$ into its traceless tensor $\phi_{\mu\nu} = h_{\mu\nu} - g^{(0)}_{\mu\nu} h/3$ and trace h. The gauge-fixed quadratic action has a wrong-sign trace mode, rotated as $h \mapsto ih$ on the perturbative contour. Gauge fixing also introduces the Faddeev–Popov determinant, the Jacobian that removes the overcounting of infinitesimally gauge-related fluctuations. It is represented by anticommuting fields, called ghosts, which form a complex Grassmann vector here. On a background with $R_{\mu\nu} = -2g_{\mu\nu}$, the relevant elliptic operators are

$$\begin{aligned} \Delta_2 &= -\nabla^2 - 2 \qquad \text{on symmetric traceless tensors,} \\ \Delta_1 &= -\nabla^2 + 2 \qquad \text{on vectors,} \\ \Delta_0 &= -\nabla^2 + 4 \qquad \text{on scalars.} \end{aligned} \tag{21}$$

The covariant Laplacian acts on the indicated tensor type. A real bosonic Gaussian contributes $(\det \Delta)^{-1/2}$, whereas a complex Grassmann Gaussian contributes $\det \Delta$. Consequently

$$Z_{1-\text{loop}} = \frac{\det \Delta_1}{\sqrt{\det \Delta_2 \det \Delta_0}}. \tag{22}$$

The asymptotically AdS boundary conditions on the thermal quotient complete the definition of these operators and their determinants. The vector numerator is essential: it subtracts the gauge directions from the metric fluctuation integral. [8, eqs. (2.25)–(2.26), (4.21)]

For an operator Δ, its heat kernel is the integral kernel of $e^{-t\Delta}$:

$$(\partial_t + \Delta_x) K(t; x, x') = 0, \qquad K(0; x, x') = \delta(x, x'), \qquad -\log\det\Delta = \int_0^\infty \frac{\mathrm{d}t}{t} \operatorname{Tr} e^{-t\Delta}. \tag{23}$$

The last formula follows by applying the proper-time representation of $-\log\lambda$ to every eigenvalue λ, with a subtraction for short-time local divergences. It converts a spectral product into a heat equation. [8, §1.1]

For a scalar of mass m_s on unit-radius $\mathbb{H}^3$, the kernel is

$$K_0(t;s) = \frac{e^{-(1+m_s^2)t - s^2/(4t)}}{(4\pi t)^{3/2}} \frac{s}{\sinh s}, \tag{24}$$

where s is geodesic distance. One checks it by using the radial Laplacian $\nabla^2 = \partial_s^2 + 2\coth s\,\partial_s$ and its short-time delta-function normalization. The factors e^{-t} and $s/\sinh s$ account for curvature. [8, eq. (2.9)] [checked]

Let g now denote the quotient generator in (11), not the metric. Linearity of the heat equation gives the image kernel

$$K_{\mathbb{H}^3/\mathbb{Z}}(t;X,X') = \sum_{n\in\mathbb{Z}} K_{\mathbb{H}^3}(t;X,g^n X'). \tag{25}$$

For tensors the isometry also acts on the indices at $g^n X'$. The $n = 0$ term is proportional to the regulated volume and produces local renormalizations. With those absorbed into the couplings and the specified vacuum energy, the interesting finite answer comes from $n \neq 0$: heat propagating between a point and its nontrivial images, or equivalently paths winding around the solid torus. [8, §§3,4.2]

The shortest distance to the nth image is $2\pi|n|y$. Changing the radial integration variable to that distance gives the measure factor proportional to $\sinh s/|\sin(\pi n\tau)|^2$. For example,

$$\int_{2\pi n y}^{\infty} \mathrm{d}s\, \sinh s\, K_0(t;s) = \frac{e^{-(1+m_s^2)t - (2\pi n y)^2/(4t)}}{4\pi^{3/2}\sqrt{t}}, \qquad n > 0. \tag{26}$$

This follows from the elementary integral $\int_a^\infty s e^{-s^2/(4t)}\mathrm{d}s = 2te^{-a^2/(4t)}$. The remaining proper-time integral then produces the scalar result

$$\log Z_{\text{scalar}} = \sum_{n=1}^{\infty} \frac{|q|^{n\Delta_s}}{n|1-q^n|^2}, \qquad \Delta_s = 1 + \sqrt{1+m_s^2}. \tag{27}$$

Here the standard AdS boundary condition selects the root $\Delta_s = 1 + \sqrt{1+m_s^2}$, with $m_s^2 \geq -1$. The result is the multiparticle partition function of a scalar primary and its global descendants. This scalar computation displays the method; no scalar matter is being added to the Einstein theory. [8, eqs. (4.5)–(4.10)] [checked]

2.3 Evaluating the graviton determinant

For gravity the index rotations in (25) give a spin-two factor $\cos(4\pi nx)$ and a spin-one ghost factor $\cos(2\pi nx)$. After the tensor, vector, and trace kernels are combined with the coefficients in (22), the nonidentity contribution is

$$\log Z_{1-\text{loop}} = \sum_{n=1}^{\infty} \frac{\sqrt{\pi}y}{2|\sin(\pi n\tau)|^2} \int_0^\infty \frac{\mathrm{d}t}{t^{3/2}} e^{-(2\pi n y)^2/(4t)} \left[e^{-t}\cos(4\pi nx) - e^{-4t}\cos(2\pi nx)\right]. \tag{28}$$

The longitudinal and scalar contributions have canceled in this combination. The remaining two terms retain the spin-dependent rotation around the quotient circle. We will see that their relative minus sign removes the unwanted lowest oscillator and gives precisely the vacuum character. [8, §2.3 and eqs. (4.22)–(4.27)]

The remaining integral can be evaluated explicitly:

$$\int_0^\infty t^{-3/2} e^{-a^2/(4t)-bt}\mathrm{d}t = \frac{2\sqrt{\pi}}{a} e^{-a\sqrt{b}}, \qquad a, b > 0. \tag{29}$$

For $a = 2\pi n y$, $b = 1, 4$, it gives

$$\log Z_{1-\text{loop}} = \sum_{n=1}^{\infty} \frac{e^{-2\pi n y}\cos(4\pi n x) - e^{-4\pi n y}\cos(2\pi n x)}{2n|\sin(\pi n \tau)|^2}. \tag{30}$$

Using $|1-q^n|^2 = 4e^{-2\pi n y}|\sin(\pi n\tau)|^2$,

$$\begin{aligned}\log Z_{1-\text{loop}} &= \sum_{n=1}^{\infty} \frac{q^{2n} + \bar{q}^{2n} - |q|^{2n}(q^n + \bar{q}^n)}{n|1-q^n|^2} \\ &= \sum_{n=1}^{\infty} \frac{1}{n}\left(\frac{q^{2n}}{1-q^n} + \frac{\bar{q}^{2n}}{1-\bar{q}^n}\right) = -\sum_{m=2}^{\infty} \log|1-q^m|^2.\end{aligned} \tag{31}$$

The second equality is a direct common-denominator identity. For the third, expand $q^{2n}/(1-q^n) = \sum_{m\geq 2} q^{mn}$ and use $-\log(1-z) = \sum_{n\geq 1} z^n/n$. The sums converge absolutely for $y > 0$. Therefore

$$Z_{1-\text{loop}} = \prod_{m=2}^{\infty} |1-q^m|^{-2}, \tag{32}$$

in agreement with the Brown–Henneaux vacuum character. [8, eqs. (4.27)–(4.28)] [checked by computation]

For the filling γ, all of this is evaluated at τ_γ. Its perturbative contribution is

$$Z_\gamma = |q_\gamma|^{-c/12} \prod_{n=2}^{\infty} |1-q_\gamma^n|^{-2}, \qquad q_\gamma = e^{2\pi i \gamma\tau}. \tag{33}$$

The one-loop factor is order c^0 in $\log Z_\gamma$ and is required before testing the spectrum. A sum of just the classical weights is a different object: in particular it misses the vacuum null-state subtraction and the threshold structure below. [6, §§2–3][8]

Exercise 2. At $x = 0$, reduce (28) to elementary exponentials and verify (32). Then check the coefficient of q^2 in the chiral factor and explain why neither a local propagating graviton nor an L_{-1} excitation is required for that coefficient to be one. [checked]

3 Regularizing the sum over fillings

3.1 Modular completion and the primary spectrum

The previous sections determine the contribution of each smooth filling. We must now combine those contributions into an observable that depends only on the boundary torus, not on the choice of basis for its cycles. A modular transformation is a large diffeomorphism of that torus. For the nonanomalous theory considered here, with $c_L = c_R$, invariance under this change of boundary marking is a required gauge symmetry of the gravitational partition function. The

vacuum contribution alone does not obey it: the S transformation turns the thermal-AdS answer into the BTZ answer. [6, §§2–3]

One way to impose the symmetry is to sum all distinct modular images of the vacuum contribution. As an algebraic procedure this averaging does not yet explain why the result should be a gravitational path integral, nor does it ensure that it counts states. Here it has two further motivations. First, the bulk classification in §3.2 identifies the images with the smooth fillings of the fixed boundary. Second, the Farey-tail examples in lecture 5 show that known string-theory indices already organize into modular images of a light-state saddle, with the appropriate measure and gauge contributions. Those examples do not prove the pure-gravity prescription, but they make it reasonable to test the analogous sum using the perturbative gravity vacuum as seed. [2, 3, 5, 6]

The resulting formal expression is

$$Z_{\text{formal}}(\tau,\bar{\tau}) = \sum_{\gamma\in\Gamma_\infty\backslash \mathrm{PSL}(2,\mathbb{Z})} Z_{\text{vac}}(\gamma\tau,\gamma\bar{\tau}). \tag{34}$$

Each term contains the classical action and boundary-graviton fluctuations of one filling. A modular transformation formally reindexes the cosets, so this expression has the required symmetry provided the divergent sum can be defined compatibly with that reindexing. The same γ acts on both chiralities because they belong to one real Euclidean metric. Consequently the sum does not factorize into independent holomorphic and antiholomorphic sums, even though the contribution of one saddle factorizes. [6, §§2–3]

A finite example: Ising and minimal-model partition functions

There are also examples involving ordinary partition functions rather than protected indices. They provide a useful test of modular averaging before we confront the infinite sum at large c. At the unitary minimal-model central charges

$$c = 1 - \frac{6}{p(p+1)}, \qquad p = 3, 4, \ldots, \tag{35}$$

Virasoro representations have additional null vectors. The vacuum contribution must therefore use the exact irreducible character $\chi^{(p)}_{\text{vac}}$, with all null states removed, in place of the $c > 1$ vacuum product. These theories have a finite vector of characters $\boldsymbol{\chi}$, transforming as $\boldsymbol{\chi}(\gamma\tau) = \rho_p(\gamma)\boldsymbol{\chi}(\tau)$. For the examples below, the orbit of $|\chi^{(p)}_{\text{vac}}|^2$ is finite. [4, §§2–3]

Let $\Gamma^{(p)}_{\text{vac}}$ be the subgroup preserving the exact vacuum contribution. It contains Γ_∞ and additional transformations that become invisible to this quantum character. The prescription is to sum each distinct function once:

$$Z^{(p)}_{\text{av}} = \sum_{\gamma\in\Gamma^{(p)}_{\text{vac}}\backslash \mathrm{PSL}(2,\mathbb{Z})} \left|\chi^{(p)}_{\text{vac}}(\gamma\tau)\right|^2. \tag{36}$$

The larger stabilizer is a quantum identification of character contributions; it does not change the classical fact that the marked solid tori form an infinite family. Since c is of order one, the AdS radius is of order the Planck length and a semiclassical expansion cannot justify this replacement on its own. Its success is tested against the exact CFT answer. [4, §§2.3,3]

For $p = 3$, the Ising CFT has $c = 1/2$ and three primaries: the identity with $h = 0$, the energy operator with $h = 1/2$, and the spin operator with $h = 1/16$. Its diagonal torus partition function

is

$$Z_{\text{Ising}} = |\chi_0|^2 + |\chi_{1/2}|^2 + |\chi_{1/16}|^2. \tag{37}$$

The modular orbit has 24 distinct terms, and its sum is

$$Z_{\text{av}}^{(3)} = 8 Z_{\text{Ising}}. \tag{38}$$

One can check this directly in the three-dimensional character space. In the order $(0, 1/2, 1/16)$, its generators are

$$S = \frac{1}{2}\begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}, \qquad T = e^{-i\pi/24}\,\mathrm{diag}(1, -1, e^{i\pi/8}). \tag{39}$$

Writing $P_0 = \mathrm{diag}(1, 0, 0)$, the images are $\boldsymbol{\chi}^\dagger \rho_p(\gamma)^\dagger P_0 \rho_p(\gamma) \boldsymbol{\chi}$. Their 24 distinct projectors sum to $8\,\mathbf{1}_3$; the trace gives $24 = 8 \times 3$. Dividing by eight fixes the vacuum multiplicity to one and reproduces the standard answer. [4, §3.2 and Appendix A] [checked by computation]

For $p = 4$, the tricritical Ising model has $c = 7/10$ and six primaries. Their weights appear in the diagonal sum

$$\begin{aligned} Z_{\text{av}}^{(4)} &= 48 Z_{\text{tri-Ising}}, \\ Z_{\text{tri-Ising}} &= \sum_{h \in \{0, 1/10, 3/5, 3/2, 3/80, 7/16\}} |\chi_h|^2. \end{aligned} \tag{40}$$

There are 288 distinct images, consistent with $288 = 48 \times 6$. Again an overall normalization gives the standard partition function. This success does not extend to every minimal-model vacuum seed. At $c = 4/5$, for example, the normalized Virasoro-vacuum sum is a weighted combination $\frac{4}{5} Z_{AA} + \frac{1}{5} Z_{AD}$ of the two minimal-model modular invariants, with fractional multiplicities. Here Z_{AA} is the diagonal invariant and Z_{AD} the three-state Potts invariant. Using an extended chiral-algebra vacuum changes the seed and allows further examples, including Potts models. [4, §§3.3,4.1,5]

The lesson is that modular images can assemble a physical spectrum, including its integer multiplicities, in concrete examples. They do not guarantee one for arbitrary vacuum data. Nor does agreement of a torus partition function by itself establish agreement of correlation functions or prove a bulk duality. We now return to the large-c vacuum seed, where the sum is infinite and its spectral interpretation has to be established by calculation. [4, §§1,3,6]

Removing the universal descendants

The state-counting test is simplest if each primary and all its Virasoro descendants are grouped into one character. Then the remaining spectral data tell us which new primaries the modular sum has generated. For $c > 1$ and a nondegenerate primary of weight $h > 0$, its character is

$$\chi_h(\tau) = q^{h - c/24} \prod_{n=1}^{\infty} (1 - q^n)^{-1} = \frac{q^{h-\xi}}{\eta(\tau)}, \qquad \xi \equiv \frac{c-1}{24}, \qquad \eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n). \tag{41}$$

The vacuum instead has $\chi_0 = q^{-\xi}(1 - q)/\eta$. Multiplying the full partition function by $\sqrt{y}\,|\eta|^2$ defines a modular invariant reduced function,

$$\mathcal{Z}(\tau, \bar{\tau}) \equiv \sqrt{y}\,|\eta(\tau)|^2 Z(\tau, \bar{\tau}). \tag{42}$$

Indeed $|\eta(\gamma\tau)|^2 = |r\tau+d|\,|\eta(\tau)|^2$ and $\sqrt{\operatorname{Im}\gamma\tau} = \sqrt{y}/|r\tau+d|$. The product is invariant. [7, §§1–3] [checked]

For a primary let

$$\delta = h - \xi, \qquad \bar\delta = \bar h - \xi, \qquad e = \delta + \bar\delta = h + \bar h - \frac{c-1}{12}, \qquad j = \delta - \bar\delta = h - \bar h. \tag{43}$$

These are the $\Delta, \bar\Delta$ of Keller–Maloney; the lowercase notation avoids confusing their sum with the total dimension of lecture 3. Physical cylinder energies are $E_L = \delta - 1/24$ and $E_R = \bar\delta - 1/24$, so $\ell M = e - 1/12$. The shift by one in $c-1$ is an exact consequence of factoring out η, not an arbitrary adjustment of the Brown–Henneaux central charge. Interpreting the new edge as a one-loop shift of the lightest black-hole mass is the bulk interpretation proposed in [7, §1].

The reduced vacuum seed is particularly simple:

$$\begin{aligned} \mathcal{Z}_{\text{vac}} &= \sqrt{y}\, q^{-\xi}\bar q^{-\xi}(1-q)(1-\bar q) \\ &= \sqrt{y}\left[q^{-\xi}\bar q^{-\xi} - q^{1-\xi}\bar q^{-\xi} - q^{-\xi}\bar q^{1-\xi} + q^{1-\xi}\bar q^{1-\xi}\right]. \end{aligned} \tag{44}$$

The minus signs remove null descendants. These four monomials are an algebraic decomposition of the vacuum character, not four physical primary states with signed multiplicities. [7, eqs. (91)–(92)] [checked]

3.2 A convergent family and its analytic continuation

A reduced seed $\sqrt{y}\, q^{\delta_*}\bar q^{\bar\delta_*}$ is invariant under T when its spin $J_s = \delta_* - \bar\delta_*$ is integral. The quotient by Γ_∞ removes this infinite repetition. There is still a divergence: as r, d grow, $y_\gamma = y/|r\tau+d|^2 \to 0$, and the modulus of a seed image behaves as $y_\gamma^{1/2} \sim |r\tau+d|^{-1}$. Summing that decay over a two-dimensional lattice does not converge. Equivalently, in (44) the factor $|1-q_\gamma|^2$ approaches a generally nonzero number on many sequences of coprime pairs, so it does not cure the divergence. [6, §3][7, §3]

Introduce a complex regulator s and define

$$P_s(\tau;\delta_*,\bar\delta_*) = \sum_{\gamma\in\Gamma_\infty\backslash \mathrm{PSL}(2,\mathbb{Z})} y_\gamma^s \exp\left(2\pi i\delta_*\gamma\tau - 2\pi i\bar\delta_*\gamma\bar\tau\right). \tag{45}$$

The asterisk labels the seed; its weights are independent of the regulator s. For $\operatorname{Re} s > 1$ the large-pair bound is $|r\tau+d|^{-2\operatorname{Re} s}$, so the sum converges absolutely. In that half-plane it is modular invariant by a legitimate reindexing. The prescription is to continue this invariant family meromorphically and evaluate at $s = 1/2$. The relevant seed combinations are regular there in the MW–KM continuation. [6, §3][7, eqs. (38)–(39)]

Writing $\mathcal{P}(\delta_*,\bar\delta_*) = P_{1/2}(\tau;\delta_*,\bar\delta_*)$ for the continued result, the candidate is

$$Z_{\text{MWK}} = \frac{1}{\sqrt{y}\,|\eta|^2}\left[\mathcal{P}(-\xi,-\xi) - \mathcal{P}(1-\xi,-\xi) - \mathcal{P}(-\xi,1-\xi) + \mathcal{P}(1-\xi,1-\xi)\right]. \tag{46}$$

MWK denotes this particular Maloney–Witten–Keller saddle sum. Analytic continuation preserves modular invariance, but it need not preserve positivity of the summands. A divergent positive series has no ordinary positive sum to which its continued value must agree. [6, 7]

3.3 Fourier expansion and the origin of the arithmetic sums

The Fourier modes in x isolate integer angular momenta. Denote seed energy and spin by $E = \delta_* + \bar{\delta}_*$ and $J_s = \delta_* - \bar{\delta}_*$. Then

$$P_s = y^s e^{-2\pi y E + 2\pi i x J_s} + \sum_{j\in\mathbb{Z}} e^{2\pi i j x} P_j(y; s, E, J_s). \tag{47}$$

The first term is the original seed; the Fourier sum contains its nonidentity images. To see how to compute it, write $d = d_0 + nr$, with d_0 a unit modulo r and $n \in \mathbb{Z}$. The identity

$$\gamma\tau = \frac{a}{r} - \frac{1}{r(r\tau + d)}, \qquad a = d_0^{-1} \pmod r \tag{48}$$

separates a phase from a function of $x + n + d_0/r$. Poisson resumming n turns the sum over these translations into a sum over integer Fourier spin j. The residual phase sum over d_0 is exactly $K(j, J_s; r)$ defined in (42) (lecture 5). [7, §3] [checked]

Expand the exponential of the remaining seed energy and spin. The result, for initially large $\operatorname{Re} s$, can be organized as

$$\begin{aligned} P_j(y; s, E, J_s) &= \sum_{m=0}^{\infty} y^{1-m-s}\, \mathcal{K}_{j,J_s}(m+s)\, \mathcal{I}_{j,m}(y; s, E, J_s), \\ \mathcal{K}_{j,J_s}(u) &= \sum_{r=1}^{\infty} \frac{K(j, J_s; r)}{r^{2u}}, \\ \mathcal{I}_{j,m} &= \frac{(2\pi)^m}{m!} \int_{-\infty}^{\infty} \mathrm{d}v\, e^{-2\pi i j y v} (1+v^2)^{-m-s} (-E - i J_s v)^m. \end{aligned} \tag{49}$$

The function $\mathcal{K}$ is a Kloosterman zeta function; v is the rescaled real integration coordinate. This supplies an explicit computational construction of (46): compute the one-dimensional Fourier integrals and arithmetic sums where they converge, then continue to $s = 1/2$. In contrast to the protected holomorphic case, the Fourier coefficient still depends nontrivially on y and must be inverse-Laplace transformed to read off energies. [7, eqs. (40)–(45)]

3.4 An explicit continuation in the zero-spin sector

The simplest calculation already displays a negative contribution. Set the seed and output spins to zero, $J_s = j = 0$. The finite Kloosterman sum is $K(0, 0; r) = \varphi(r)$, where φ is Euler's totient function, the number of units modulo r. Its Dirichlet series is

$$\sum_{r=1}^{\infty} \frac{\varphi(r)}{r^{2u}} = \frac{\zeta(2u-1)}{\zeta(2u)}. \tag{50}$$

The Fourier integral follows from $\int_{-\infty}^{\infty} (1+v^2)^{-a} \mathrm{d}v = \sqrt{\pi}\,\Gamma(a - 1/2)/\Gamma(a)$. Thus the mth term of the zero Fourier mode is

$$P_{0,m} = y^{1-m-s} \frac{\zeta(2m + 2s - 1)}{\zeta(2m + 2s)} \frac{2^m \pi^{m+1/2} \Gamma(m + s - 1/2)}{m!\,\Gamma(m+s)} (-E)^m. \tag{51}$$

[7, eq. (47)] [checked]

For $m = 0$, the gamma function has a pole at $s = 1/2$, but the reciprocal zeta function has a zero there. Setting $s = 1/2 + \epsilon$,

$$\Gamma(\epsilon) \sim \epsilon^{-1}, \qquad \zeta(1 + 2\epsilon) \sim (2\epsilon)^{-1}, \qquad \zeta(2\epsilon) \longrightarrow \zeta(0) = -\tfrac{1}{2}. \tag{52}$$

The factors combine to give

$$P_{0,0}(y; 1/2, E, 0) = -\sqrt{y}. \tag{53}$$

For $m \geq 1$ no such pole remains, and

$$P_{0,m}(y; 1/2, E, 0) = \frac{\zeta(2m)}{\zeta(2m+1)} \frac{2^m \pi^{m+1/2}}{m\,\Gamma(m+1/2)} (-E)^m y^{1/2-m}. \tag{54}$$

The finite negative answer in (53) is a consequence of the continuation, specifically $\zeta(0) = -1/2$. No negative elementary oscillator has been introduced. [7, eqs. (48)–(49)] [checked]

For later use, the same $m = 0$ zero-mode calculation with $J_s = \pm 1$ has $K(0, \pm 1; r) = \mu(r)$, the Möbius function, and $\sum_r \mu(r) r^{-2s} = 1/\zeta(2s)$. The integral at $m = 0$ is unchanged, so its continuation is instead

$$P_{0,0}(y; 1/2, E, \pm 1) = 2\sqrt{y}. \tag{55}$$

Together these two elementary arithmetic continuations will produce the -6 of the vacuum sum. [6, eqs. (3.35)–(3.38)] [checked]

Exercise 3. Use the Euler products of ζ to derive the totient and Möbius Dirichlet-series identities above. Explain why substituting $s = 1/2$ in the individual divergent factors of (51) is invalid, whereas taking their combined limit gives (53). [checked]

4 Extracting the spectrum

4.1 Censored states, spin quantization, and a continuous energy density

We have obtained a finite modular function, but its Fourier expansion is not yet a list of energy levels. A Fourier mode in x fixes the spin; within that mode, the dependence on y is a Boltzmann transform of the energies. Inverting this transform reveals both the support and the sign of the primary spectral measure. For one seed, define the density of the additional primaries by

$$y^{-1/2} P_j(y; 1/2, E, J_s) = \int_{-\infty}^{\infty} \mathrm{d}e\, \rho_{E,J_s}(e, j) e^{-2\pi y e}. \tag{56}$$

After computing its support, the integral will run over $e \geq |j|$. Dividing by $|\eta|^2$ subsequently restores the Virasoro descendants. The density here counts primaries; the original seed is kept separately. [7, §4]

To compare that support with the black-hole region, call a primary censored if

$$\delta < 0 \quad \text{or} \quad \bar{\delta} < 0, \qquad \text{equivalently} \quad h < \xi \quad \text{or} \quad \bar{h} < \xi. \tag{57}$$

The uncensored region has $\delta, \bar{\delta} \geq 0$, or $e \geq |j|$. It is the two-sided BTZ bound after the descendant shift. A bound on $h + \bar{h}$ alone would not express it: a state can have arbitrarily large h while $\bar{h}$ remains below the threshold. To measure the distance from a rotating black-hole bound, it is useful to define the *twist*, the dimension minus the magnitude of the spin:

$$\mathfrak{t} \equiv h + \bar{h} - |j| = 2\min(h, \bar{h}), \qquad \ell M - |J| = \mathfrak{t} - \frac{c}{12}. \tag{58}$$

The second equality uses the cylinder charges from our conventions. Classical extremality corresponds to $\mathfrak{t} = c/12$; the reduced primary edge is $\mathfrak{t} = 2\xi = (c-1)/12$. Twist isolates the less excited chirality and therefore remains small for highly energetic states close to extremality. A twist gap is the smallest twist among nonvacuum primary operators, or the infimum if it is not attained. We will use this variable in §7.4 to examine the spectrum near the edge at large spin. The charged polar condition in lecture 5 refers instead to the charge-subtracted energy of an index. [7, §1][14, §§2–4] [checked]

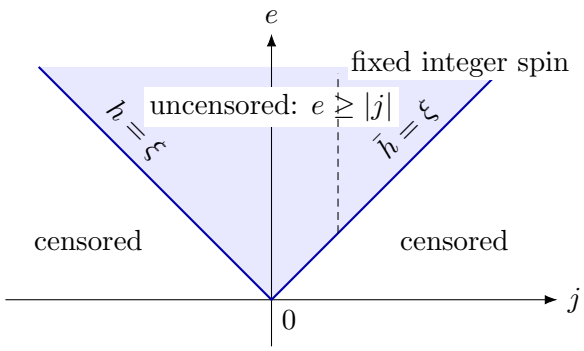


Figure 3: Reduced energy and spin, as defined in (43). The nonidentity Poincaré images have support on the wedge $e \geq |j|$. At each integer spin the spectrum from the sum is continuous in e. The drawing illustrates the support derived below, following [7].

The zero-spin inverse transform is elementary:

$$\int_0^\infty \frac{(2\pi)^m e^{m-1}}{\Gamma(m)} \exp(-2\pi y e) \mathrm{d}e = y^{-m}, \qquad m > 0. \tag{59}$$

Equations (53)–(54) therefore give, for a scalar seed,

$$\begin{aligned} \rho_{E,0}(e,0) &= -\delta_{\mathrm{D}}(e) + \sum_{m=1}^\infty 2^{m-1} A_m \frac{\zeta(2m)}{\zeta(2m+1)} (-E)^m e^{m-1} \Theta(e), \\ A_m &= \frac{2^{3m+1}\pi^{2m}}{(2m)!}. \end{aligned} \tag{60}$$

The symbol δ_{D} denotes a Dirac delta function; Θ is the step function. The conversion of gamma functions uses $\Gamma(m+1/2) = (2m)!\sqrt{\pi}/(4^m m!)$. The $m \geq 1$ terms are ordinary functions on a continuum of positive energies. For a scalar seed with $E < 0$, every one of their coefficients is positive. The sole negative term of this particular seed is the atom $-\delta_{\mathrm{D}}(e)$. [7, eqs. (71)–(72)] [checked]

For $j \neq 0$ and a scalar seed, the Fourier integral instead produces $K_m(2\pi y|j|)$, the modified Bessel function of the second kind. Its integral representation

$$K_m(a) = \int_0^\infty e^{-a\cosh u} \cosh(mu) \mathrm{d}u = \int_1^\infty e^{-at} \frac{\cosh(m \operatorname{arcosh} t)}{\sqrt{t^2-1}} \mathrm{d}t \tag{61}$$

shows the support of the inverse Laplace transform without an asymptotic approximation. Define

$$f_m(t) = \frac{\cosh(m \operatorname{arcosh} t)}{\sqrt{t^2-1}} \Theta(t-1). \tag{62}$$

The continued series gives

$$\rho_{E,0}(e,j)=\sum_{m=1}^{\infty}\frac{\sigma_{2m}(|j|)}{|j|^{2m}\zeta(2m+1)}A_m|j|^{m-1}(-E)^m f_m(e/|j|),\qquad j\neq 0. \tag{63}$$

Each term vanishes for $e<|j|$ and is positive above the edge if $E<0$. The inverse transformation involves an interval of energies, not a set of isolated levels. [7, eqs. (73)–(74)] [checked]

For a spinning seed, the polynomial $(-E-iJ_s v)^m$ in (49) introduces derivatives of Bessel functions. Keller–Maloney perform these transforms in §4.3 of [7]; the same support condition survives:

$$\rho_{E,J_s}(e,j)=0\qquad\text{for}\quad e<|j|. \tag{64}$$

Thus the nonidentity images add no censored primaries. Their signs, however, are more delicate than their support, especially after the four-term vacuum subtraction. These results supply a useful modular construction with prescribed censored data, even when it does not supply the spectrum of a compact unitary CFT. [7, §§1,4]

4.2 A modular transform as an integral over energies

The appearance of a continuum can also be understood directly from a single modular transform. Treat a nondegenerate character as $\chi_\delta(\tau)=q^\delta/\eta(\tau)$, with the subscript now indicating its reduced weight. The Gaussian integral gives

$$\chi_\delta(-1/\tau)=\int_0^\infty \mathrm{d}\delta'\,\frac{\sqrt{2}}{\sqrt{\delta'}}\cos\!\big(4\pi\sqrt{\delta\delta'}\big)\chi_{\delta'}(\tau). \tag{65}$$

For $\delta<0$ the cosine is continued to a hyperbolic cosine. To verify the normalization, set $\delta'=P^2$ and use

$$2\sqrt{2}\int_0^\infty \mathrm{d}P\,e^{2\pi i\tau P^2}\cos(4\pi\sqrt{\delta}P)=\frac{1}{\sqrt{-i\tau}}e^{-2\pi i\delta/\tau}, \tag{66}$$

then use $\eta(-1/\tau)=\sqrt{-i\tau}\eta(\tau)$. The Gaussian converges for $\operatorname{Im}\tau>0$. This is the character version of the continuous S-transform construction discussed in [7, §2]. [checked]

The two chiral kernels give a continuous density on $\delta',\bar\delta'\geq 0$. A single S image does not yet enforce integer spin. The full Poincaré sum includes the translations and arithmetic phases that discretize the spin, but the energy integral remains continuous, as (60) and (63) establish explicitly. Holomorphic Rademacher reconstruction had a different constraint: integer powers of a single q in a meromorphic modular function. There is no corresponding holomorphy argument that forces the nonholomorphic answer to become an energy lattice. [7, §§2–4]

The vacuum is the difference of two nondegenerate characters in each chirality. For $c>25$, its chiral S kernel is

$$S_{\mathrm{vac}}(\delta')=\frac{\sqrt{2}}{\sqrt{\delta'}}\left[\cosh(4\pi\sqrt{\xi\delta'})-\cosh(4\pi\sqrt{(\xi-1)\delta'})\right]. \tag{67}$$

It is positive for $\delta'>0$ and behaves as $8\sqrt{2}\pi^2\sqrt{\delta'}$ at the edge. The cancellation of the constant terms is another expression of the missing L_{-1} state. Other modular images have different arithmetic phases; their cancellations need not match this one. This will matter for the near-extremal large-spin spectrum. [14, 15] [checked]

At fixed $c > 25$, take both reduced chiral energies $\delta = (e+j)/2$ and $\bar{\delta} = (e-j)/2$ large. The vacuum kernel then gives

$$\log \rho_{\text{prim}}(e,j) \sim 4\pi\sqrt{\frac{\xi(e+j)}{2}} + 4\pi\sqrt{\frac{\xi(e-j)}{2}}. \tag{68}$$

This asymptotic is not uniform when either reduced chiral energy approaches zero. It is Cardy growth for primaries with the descendant shift; restoring descendants gives the usual total-state Cardy growth governed by c. At leading semiclassical order the difference between $c-1$ and c is subleading, and the exponent reproduces the BTZ entropy. Positive high-energy asymptotics are therefore compatible with defects in the exact low-energy or near-extremal spectral measure. [7, §4.5]

5 The failures of the vacuum saddle sum

The modular completion has passed several necessary tests: it has the correct boundary symmetry, reproduces the classical thermal saddles, and retains the perturbative vacuum module. These tests concern its transformation law and semiclassical expansion. To interpret it as an exact thermal trace, we must also examine the spectral measure extracted in §4. A positive discrete measure with integer weights would count primary states; a negative contribution cannot, and a smooth continuum would require a different interpretation of the observable. We begin at the scalar threshold, where the failure is already visible in the first term of the low-temperature expansion, and then examine the more delicate limit of large spin near extremality. [6, 7, 14, 15]

5.1 The negative threshold contribution

The vacuum is not a single unrestricted Verma module: its level-one null states must be removed. We therefore have to combine the four seed densities before testing positivity. Using (44), with energies and spins $(E, J_s) = (-2\xi, 0), (-2\xi+1, 1), (-2\xi+1, -1), (-2\xi+2, 0)$, gives

$$\rho_{\text{vac}}(e,j) = \rho_{-2\xi,0}(e,j) - \rho_{-2\xi+1,1}(e,j) - \rho_{-2\xi+1,-1}(e,j) + \rho_{-2\xi+2,0}(e,j). \tag{69}$$

The $m=0$, $j=0$ terms are fixed by (53) and (55): $-1-2-2-1=-6$. Therefore

$$\rho_{\text{vac}}(e,0) = -6\delta_{\text{D}}(e) + \rho_{\text{vac}}^{\text{cont}}(e,0). \tag{70}$$

The negative atom is located at

$$h = \bar{h} = \xi = \frac{c-1}{24}, \qquad E_L = E_R = -\frac{1}{24}, \qquad \ell M = -\frac{1}{12}. \tag{71}$$

It is slightly below the classical massless BTZ energy in the physical cylinder convention. In the reduced primary variables it is exactly at the edge $e=0$. [6, eq. (3.39)][7, eq. (93)] [checked]

To check whether the continuous contribution could compensate this negative atom, retain the remaining zero-spin terms:

$$\rho_{\text{vac}}^{\text{cont}}(e,0) = \Theta(e) \sum_{m=1}^{\infty} \frac{2^{m-1} A_m e^{m-1}}{\zeta(2m+1)} \left\{ \zeta(2m) \left[(2\xi)^m + (2\xi-2)^m \right] - 2^{2-m} T_m(2\xi-1) \right\}, \tag{72}$$

where T_m is the Chebyshev polynomial defined by $T_m(\cosh u) = \cosh(mu)$. This scalar continuum is positive for $\xi \geq 1/2$, or $c \geq 13$. [7, eq. (94)] The continuous positive terms cannot cancel a

negative delta function: integrating against a nonnegative test function supported in a sufficiently small neighborhood of the edge retains the negative atom. Positivity of a spectral distribution must hold for such tests, not just pointwise away from the edge.

The same result is visible directly at low temperature:

$$Z_{\mathrm{MWK}} = Z_{\mathrm{vac}} + \frac{1}{|\eta(\tau)|^2}\left[-6 + O(y^{-1})\right], \qquad y \longrightarrow \infty. \tag{73}$$

The inverse powers arise from (54); nonzero spin modes are exponentially suppressed in this limit. It is an exact spectral obstruction to interpreting this particular regularized answer as a trace over a compact unitary CFT Hilbert space, despite its correct classical action and boundary-graviton spectrum. [6, §3.3][7, §4.4]

5.2 The continuous spectrum and the state-counting test

Removing the negative threshold term would satisfy only one part of the state-counting requirement. The inverse transforms in the previous section also give ordinary functions of energy on an interval. To understand why this differs from the expected spectrum, compare the measures themselves, rather than only their integrated high-energy growth. [6, §2.3][7, §4]

There is an independent problem even after the atom is removed. At a fixed spin a compact CFT would have

$$\rho_{\mathrm{CFT}}(e, j) = \sum_{\alpha:\, j_\alpha = j} d_\alpha \delta_{\mathrm{D}}(e - e_\alpha), \qquad d_\alpha \in \mathbb{Z}_{\geq 0}. \tag{74}$$

The functions in (60)–(63) are not of this form. If a discrete spectrum has an isolated next energy above the threshold, the corresponding correction to a low-temperature trace is exponentially small in y. A continuum with density proportional to e^{m-1} at the edge instead gives y^{-m}, exactly as in (59). This explains the power-law diagnostic in Maloney–Witten's analysis. [6, §2.3 and eq. (3.39)][7, §4]

A noncompact CFT can have a continuous spectrum; a semiclassical coarse-grained density can also be continuous. Neither possibility is the proposed exact partition function of a single compact CFT with the vacuum and finite degeneracies assumed in (7). Moreover neither makes a negative spectral measure acceptable. A coarse-grained or averaged interpretation of the smooth-saddle answer requires extra input about the observable being computed. [7, §§1,5]

5.3 Keller–Maloney's modular correction at the edge

The negative atom can be canceled without changing the censored primary spectrum. Keller–Maloney give a simple modular invariant building block, the self-dual compact-boson partition function,

$$Z_{\mathrm{bos}}(\tau, \bar{\tau}) = \frac{1}{|\eta(\tau)|^2} \sum_{a,b \in \mathbb{Z}} q^{(a+b)^2/4} \bar{q}^{(a-b)^2/4}. \tag{75}$$

Its modular invariance follows from the even self-dual momentum-winding lattice; it also follows by Poisson resumming the two lattice sums. When used as a correction to a theory at central charge c, the terms are read as primary characters with

$$h = \xi + \frac{(a+b)^2}{4}, \qquad \bar{h} = \xi + \frac{(a-b)^2}{4}, \qquad j = ab \in \mathbb{Z}. \tag{76}$$

They are all uncensored. The $(a,b)=(0,0)$ term contributes $+\delta_{\mathrm{D}}(e)$ at zero spin. Hence

$$Z_{\text{edge corrected}} = Z_{\mathrm{MWK}} + 6Z_{\mathrm{bos}} \tag{77}$$

cancels the -6 atom. A single scalar-seed Poincaré series only has the -1 atom and needs one copy. [7, §§4.2,4.4–5][15, §2] [checked]

This is an addition of modular functions at the same physical central charge, with the weights shifted as in (76). It is not tensoring the hypothetical gravity CFT with six bosons, which would multiply the partition function and change the central charge. It adds a complete modular invariant lattice spectrum containing the threshold primary. Modular invariance does not single out this particular completion. Its positive discrete contributions leave the continuous part of the MWK density in place. Thus the cancellation of the atom does not provide a discrete spectrum, operator products, or a construction of a dual CFT. [7, §§4.2,5] [checked]

5.4 Negative density at large spin near extremality

The scalar threshold is not the only obstruction to positivity. The large-c estimate in the original analysis suggested that, after the -6 term was removed, the remaining vacuum density could be positive. The difficulty is that this estimate is not uniform across the spectrum: a term subleading at a fixed distance from the black-hole threshold can dominate arbitrarily close to it. The lightcone modular-bootstrap analysis of Benjamin–Ooguri–Shao–Wang exposed this problem; the modular-kernel analysis of Benjamin–Collier–Maloney makes its origin particularly explicit. We now derive the negative contribution and the region in which it wins. The modifications of the light spectrum proposed in these works will be considered in a later lecture. [7, §4.4][14, §§4–6][15, §2]

The useful variable is the distance above the edge at fixed spin, not the total energy. Recall the twist $\mathfrak{t}=h+\bar{h}-|h-\bar{h}|=2\min(h,\bar{h})$ defined in §4. For a large positive spin, increasing h alone raises both the energy and angular momentum while leaving $\bar{h}$ small. Such states can have very large total energy and still lie close to extremality. In the reduced variables define

$$t=\min(h,\bar{h})-\xi=\frac{e-|j|}{2}=\frac{\mathfrak{t}-2\xi}{2},\qquad \varrho_j(t)=2\rho_{\mathrm{vac}}(j+2t,j),\quad j>0. \tag{78}$$

The factor of two follows from $\mathrm{d}e=2\mathrm{d}t$: $\varrho_j(t)\mathrm{d}t$ is the primary spectral measure in this variable. The continuum starts at $t=0$, the quantum-shifted primary edge. Classical BTZ extremality, $\min(h,\bar{h})=c/24$, instead lies at $t=1/24$. Thus large spin near the primary edge means $j\to\infty$ and $t\to 0^+$; it does not mean approaching the scalar point $e=j=0$. [14, §4][15, §2.1] [checked]

To follow the cancellation responsible for the sign, retain the individual modular denominators r. Work at fixed $c>25$, so $\xi>1$, and abbreviate the two chiral kernel factors as

$$a_r(v)=\cosh\left(\frac{4\pi}{r}\sqrt{\xi v}\right),\qquad b_r(v)=\cosh\left(\frac{4\pi}{r}\sqrt{(\xi-1)v}\right),\qquad v\geq 0. \tag{79}$$

The first arises from the vacuum weight $h=0$, the second from the subtraction at $h=1$. Combining the four seeds gives

$$\varrho_j^{(r)}(t)=\frac{2}{r\sqrt{t(t+j)}}\Big\{K(j,0;r)\big[a_r(t+j)a_r(t)+b_r(t+j)b_r(t)\big] \\ -K(j,-1;r)a_r(t+j)b_r(t)-K(j,1;r)b_r(t+j)a_r(t)\Big\}. \tag{80}$$

The Kloosterman sums K were defined in (42) (lecture 5). Their second arguments are the spins $0, -1, 1$ of the vacuum seed and its null-state subtractions. The continued sum over r gives the MWK continuum, with the same regularization discussed earlier. Here r labels a modular denominator, not an order in perturbation theory. Writing the answer this way lets us see how the arithmetic phases modify the vacuum cancellation. [15, eqs. (2.5)–(2.8) and Appendix A]

For $r = 1$ all three Kloosterman sums equal one, so the expression factorizes into two positive differences:

$$\varrho_j^{(1)}(t) = \frac{2}{\sqrt{t(t+j)}}[a_1(t+j) - b_1(t+j)]\,[a_1(t) - b_1(t)]. \tag{81}$$

Near $t = 0$, the second bracket is $8\pi^2 t + O(t^2)$: its constant terms cancel because the level-one vacuum state is null. At large j and fixed c, the first bracket has leading behavior $\frac{1}{2}e^{4\pi\sqrt{\xi j}}$. Together they give

$$\varrho_j^{(1)}(t) \sim A_j\sqrt{t}, \qquad A_j = \frac{8\pi^2}{\sqrt{j}}e^{4\pi\sqrt{\xi j}}. \tag{82}$$

This is the same suppression at the edge that we found from the single S kernel in (67). The large black-hole exponential is multiplied by a factor that vanishes at threshold. [15, eq. (2.12)] [checked]

For $r = 2$ the only invertible residue is $d = 1$, with inverse one. The phases can therefore be evaluated without approximation:

$$K(j, 0; 2) = (-1)^j, \qquad K(j, \pm 1; 2) = -(-1)^j. \tag{83}$$

The extra minus sign on the spin-one seeds turns the differences into sums:

$$\varrho_j^{(2)}(t) = \frac{(-1)^j}{\sqrt{t(t+j)}}[a_2(t+j) + b_2(t+j)]\,[a_2(t) + b_2(t)]. \tag{84}$$

The second bracket now tends to two, so the edge suppression is lost. At large spin,

$$\varrho_j^{(2)}(t) \sim \frac{(-1)^j B_j}{\sqrt{t}}, \qquad B_j = \frac{1}{\sqrt{j}}\left[e^{2\pi\sqrt{\xi j}} + e^{2\pi\sqrt{(\xi-1)j}}\right]. \tag{85}$$

For odd j this contribution is negative. It has a smaller exponential than the $r = 1$ term, but it diverges as $t^{-1/2}$ instead of vanishing as $t^{1/2}$. The competition is therefore controlled by both the large-spin exponential and the distance to the edge. [14, §4][15, eq. (2.12)] [checked]

For large odd spin, the two leading contributions give $\varrho_j(t) \simeq A_j\sqrt{t} - B_j/\sqrt{t}$. Define the crossover scale $t_j = B_j/A_j$. At fixed $c > 25$ and $j \to \infty$,

$$t_j \sim \frac{1}{8\pi^2}e^{-2\pi\sqrt{\xi j}}, \qquad 0 < t \ll t_j \quad \Longrightarrow \quad \varrho_j(t) < 0. \tag{86}$$

The higher-denominator terms have smaller large-spin exponentials and do not reverse this sign in the stated limit. At a fixed positive t, the $r = 1$ term eventually dominates. In the shrinking interval (86), however, the $r = 2$ term dominates. This explains why an approximation controlled away from the edge cannot establish positivity everywhere. [14, §§4–6][15, §2]

There are negative intervals at arbitrarily large odd spins, so this failure involves infinitely many spin sectors. Each interval has a continuous set of energies with negative density; it cannot

be represented as a finite collection of negative multiplicities at the scalar threshold. The singularity $t^{-1/2}$ is integrable at fixed spin, but even the integrated negative weight need not become small as the interval narrows. For fixed $0 < \varepsilon < 1$, integrating the two leading terms gives

$$\int_0^{\varepsilon t_j} \varrho_j(t)\mathrm{d}t \simeq 2B_j\sqrt{t_j}\left(\frac{\varepsilon^{3/2}}{3} - \sqrt{\varepsilon}\right) < 0, \qquad B_j\sqrt{t_j} \;\propto\; j^{-1/2}e^{\pi\sqrt{\xi j}}. \tag{87}$$

The proportionality is the fixed-c, large-j asymptotic following from (82)–(85). Thus an exponentially narrow window can still carry a large negative spectral weight. This is a statement about the computed density, not a claim that a Hilbert space with those states has been constructed. [15, §2] [checked]

The lightcone modular bootstrap reaches the same problematic regime from boundary consistency. Continuing the left and right inverse temperatures independently allows one chirality to probe high energies while the other remains near its lowest primary energy. Modular crossing then relates this limit to the vacuum in a transformed channel. Expansions associated with different rational cusps resolve more than the leading Cardy growth: the denominator-two channel retains the $(-1)^j$ phase that distinguishes odd and even spins. Under the vacuum-dominance assumptions used in that analysis, the resulting universal contribution has the sign mechanism just derived. For the specific MWK sum, the negative density is independently established by its explicit modular kernels; extending the conclusion to a bound on the spectrum of every CFT requires additional control of nonvacuum contributions and limits. [14, §§3–5][15, §2]

Finally, the compact-boson correction in (77) supplies positive delta functions at a discrete set of energies. It removes the scalar atom, but cannot cancel a negative continuous density on these open intervals: a nonnegative test function supported away from its atoms still has a negative integral. The later problem therefore remains after that partial correction. Its resolution requires more information than the vacuum and its smooth modular images; we postpone the proposed additional states and bulk configurations to the lectures devoted to those completions. [15, §2]

5.5 What modular invariance leaves undetermined

The contrast with lecture 5 is now explicit. Holomorphy on the modular curve, together with a specified cusp pole, made the extremal partition function a finite polynomial problem. A nonholomorphic modular invariant is a function on a two-real-dimensional space, and its low-energy data do not determine all of it. Poincaré sums provide particular completions of censored spectra. One may add other modular invariant functions with no censored support, including (75), without changing those prescribed light states. More general modular functions can be organized by the spectral theory of the hyperbolic Laplacian, including Maass cusp forms and Eisenstein series; they need not have a positive CFT spectral interpretation. [7, §§2,5 and Appendix A]

This freedom does not mean that every change can be called a regularization of the same path integral. In §5 of [7], the authors demand modular invariance, no new censored states, and preservation of a convergent Laplacian-derived observable of the original sum. Under these additional conditions the allowed change is a tightly constrained homogeneous solution, insufficient to remove the spectral defects. The compact-boson completion relaxes those requirements on the original sum and specifies additional spectral data. The choice of a completion therefore needs physical input beyond modular invariance.

Nor is rounding the smooth density to integers an exact solution. When exponentially many states occupy a macroscopic energy window, a continuum can approximate their count very accurately; an individual level spacing can be exponentially small in the entropy. Recovering a

delta-function spectrum from that approximation needs information about all level locations and multiplicities. Arbitrary rounding has no reason to preserve the exact modular crossing relations. Keller–Maloney discuss a possible discrete completion through changes subleading in the large-c expansion, but the Poincaré construction does not exhibit the operator algebra of such a CFT, and the later near-extremal result restricts the proposed minimal spectrum further. [7, §5][14]

The established outcome is therefore specific. The regularized sum of smooth solid-torus saddles has the expected modular covariance, classical BTZ thermodynamics, and perturbative boundary gravitons. Its added primary spectrum lies in the uncensored wedge and displays black-hole growth at high energy. Its exact spectral measure also contains a negative threshold atom, a further negative near-extremal region at odd spin, and a continuous energy distribution. These facts exclude this saddle sum as the ordinary torus trace of a compact unitary CFT. They do not by themselves classify all possible quantum completions of the Einstein theory. A completion must specify the extra states, configurations, or observable and demonstrate that the resulting CFT consistency conditions are met. [6, 7, 8, 14, 15]

Exercise 4. Rewrite one lattice term of (75) as a product of nondegenerate $c > 1$ Virasoro characters, and verify its spin and threshold. Explain separately why adding six copies removes (70), why it does not change censored data, and why it cannot make (60) discrete. [checked]

Exercise 5. At large odd spin, compare (82) and (85). Derive the exponential scale in (86). Then determine which order of limits, $t \to 0$ or $c, j \to \infty$, would conceal the negative region. This provides a concrete test of uniformity in a semiclassical estimate. [14, 15]

A Checks on conventions and spectral formulas

The following short checks collect relations that otherwise look different in the main sources.

Levels and physical energies. In the vector-trace normalization $k = c/24 = \ell/(16G)$, while lecture 2 uses $k_{\mathrm{CS}} = 4k$. The vacuum has $\ell M = -c/12 = -2k$. The reduced energy used in the primary transform is $e = \ell M + 1/12$; hence $e = 0$ in (70) means $\ell M = -1/12$, independently of c. These are identities from (1), (2), and (43). [checked]

The first continuous scalar coefficient. For $m = 1$, $A_1 = 8\pi^2$. Equation (72) gives

$$\rho_{\mathrm{vac}}^{\mathrm{cont}}(e, 0) = \frac{8\pi^2}{\zeta(3)}(4\xi - 2)\big(\zeta(2) - 1\big) + O(e). \tag{88}$$

Thus for $c > 13$ the continuum begins with a positive constant. Its Laplace transform yields

$$Z_{\mathrm{MWK}} - Z_{\mathrm{vac}} = \frac{1}{|\eta|^2}\left[-6 + \frac{4\pi(4\xi - 2)(\zeta(2) - 1)}{\zeta(3)y} + O(y^{-2})\right], \tag{89}$$

up to exponentially small nonzero-spin terms. Substituting $\xi = k - 1/24$ gives the coefficient $(\pi^3 - 6\pi)(24k - 13)/(9\zeta(3))$. The printed [6, eqs. (3.7), (3.47)] instead yields $24k + 11$: in its four-seed decomposition the shifts of κ appear as $+1, +2$. Directly multiplying $|q|^{-2\xi}$ by $(1 - q)(1 - \bar{q})$ shows that these shifts must be $-1, -2$. Equations (92)–(94) of [7] have the latter, consistent signs, and give the coefficient used here. The threshold -6 and the continuous-spectrum diagnosis are unaffected by this discrepancy. [checked by computation]

The scalar kernel and the proper-time integral. For $s > 0$, differentiating (24) verifies $(\partial_t - \partial_s^2 - 2\coth s\,\partial_s + m_s^2)K_0 = 0$. Its $s \to 0$, $t \to 0$ limit reduces to the flat three-dimensional Gaussian normalization. Differentiating the standard Gaussian integral with respect to a, or making a change of variable $t = a^2/(4v^2)$, yields (29). The algebraic passage from the winding sum to (32) was checked independently at nonzero x, so the agreement is not restricted to the nonrotating torus. [checked by computation]

Version-sensitive conclusions. The extremal proposal is described using 0706.3359v1; the Poincaré construction and proposed threshold repair use 1407.6008v1. The corrected Farey-tail account uses 0712.0573 and the correction in version 3 of hep-th/0005003. The positivity claim of the 2014 vacuum analysis is qualified using the explicit 2019–2020 near-extremal calculation. The Lin–Pei exclusions are stated with their conjectural TMF input, and Gaiotto–Yin's genus-two consistency tests are not labelled nonexistence theorems. These qualifications are part of the physical conclusions, not just changes of notation. [1, 7, 3, 2, 14, 15, 12, 10, 11]

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